

WOLF POINT CLAYTON, MT, USA

WMO: 727686

Lat: 48.094N

Lon: 105.574W

Elev: 605

StdP: 94.26

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94017

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-26.1	-32.5	0.2	-29.0	-29.2	0.3	-25.5	12.7	-4.3	11.4	-4.7	2.3	250	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	35.3	18.9	33.3	18.7	31.3	18.2	21.5	30.4	20.3	29.2	19.3	28.4	4.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.5	26.0	17.4	13.4	23.9	16.2	12.4	22.8	65.5	30.3	60.9	28.9	57.3	28.3	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.3	8.8	-34.3	38.5	3.3	1.8	-36.6	39.7	-38.5	40.8	-40.3	41.8	-42.7	43.1		
(5)			-33.8	23.7	2.5	1.5	-35.6	24.8	-37.1	25.7	-38.5	26.5	-40.3	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-9.9	-9.1	-0.9	6.5	12.2	17.4	22.1	20.7	14.6	6.8	-1.6	-8.7	(6)
(7)		DBStd	12.67	7.89	7.84	7.55	4.97	4.64	3.70	3.33	3.62	4.62	4.97	6.75	7.43	(7)
(8)		HDD10.0	2709	617	535	341	124	31	1	0	0	12	121	348	580	(8)
(9)		HDD18.3	4801	875	768	597	356	197	60	7	18	130	358	597	838	(9)
(10)		CDD10.0	1219	0	0	2	19	98	222	375	332	149	22	1	0	(10)
(11)		CDD18.3	268	0	0	0	0	6	31	123	91	17	1	0	0	(11)
(12)		CDH23.3	4394	0	0	1	29	179	559	1759	1415	419	32	0	0	(12)
(13)		CDH26.7	1980	0	0	0	5	44	193	867	689	174	7	0	0	(13)
(14)	Wind	WSAvg	3.6	3.4	3.3	3.9	4.5	4.3	3.8	3.5	3.4	3.3	3.6	3.3	3.3	(14)
(15)	Precipitation	PrecAvg	356	10	6	12	25	53	82	53	43	25	21	10	8	(15)
(16)		PrecMax	499	36	18	37	63	141	164	156	180	87	84	79	21	(16)
(17)		PrecMin	162	1	0	0	7	10	30	11	1	2	0	0	0	(17)
(18)		PrecStd	75	8	4	9	16	33	36	34	41	23	17	15	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	11.2	21.9	27.0	30.7	34.1	38.0	37.8	34.6	27.4	18.6	9.3	(19)
(20)			MCWB	4.2	5.3	10.7	13.0	16.6	18.7	19.7	19.5	18.2	14.1	9.7	4.0	(20)
(21)		2%	DB	4.8	6.7	17.1	22.7	27.7	31.2	35.9	35.0	31.3	22.4	14.1	5.9	(21)
(22)			MCWB	1.9	2.7	8.3	11.5	15.3	18.9	19.7	18.5	16.7	12.3	7.2	2.3	(22)
(23)		5%	DB	2.8	3.7	13.0	19.7	25.3	28.8	33.8	32.9	28.2	18.9	11.0	3.2	(23)
(24)			MCWB	0.8	1.2	6.2	9.8	14.2	17.3	19.3	18.1	15.8	10.5	5.6	0.8	(24)
(25)		10%	DB	1.1	1.6	9.8	17.2	22.6	26.7	31.8	30.9	25.0	16.1	7.7	1.3	(25)
(26)			MCWB	-0.6	-0.1	4.8	8.6	12.9	16.7	18.9	17.7	14.5	9.1	3.7	-0.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	5.5	10.8	13.9	18.2	22.1	24.0	21.9	19.9	15.4	10.2	4.3	(27)
(28)			MCD B	8.0	10.5	20.9	24.5	26.8	29.9	32.8	32.9	28.6	25.4	17.3	8.2	(28)
(29)		2%	WB	2.2	3.0	8.9	11.8	16.4	20.2	22.2	20.3	18.0	13.0	7.8	2.4	(29)
(30)			MCD B	4.5	6.2	16.7	21.6	25.2	28.2	30.8	30.1	28.1	20.1	14.0	5.4	(30)
(31)		5%	WB	0.9	1.4	6.7	10.3	15.1	18.8	20.9	19.4	16.5	11.1	5.7	1.0	(31)
(32)			MCD B	2.8	3.7	12.6	18.5	23.6	26.4	29.5	29.6	26.5	18.1	10.3	3.1	(32)
(33)		10%	WB	-0.5	-0.1	5.0	9.1	13.7	17.5	19.9	18.5	15.1	9.7	4.0	-0.4	(33)
(34)			MCD B	1.1	1.7	9.3	16.2	21.1	24.8	28.7	28.6	24.2	15.7	7.5	1.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	12.1	13.0	15.0	15.0	14.5	16.5	17.0	16.7	14.6	13.2	11.5	(35)
(36)			MCDBR	12.7	14.7	19.0	21.0	19.5	18.9	19.8	20.9	22.4	20.7	18.4	14.0	(36)
(37)			MCWBR	10.3	11.3	11.4	11.1	9.4	8.4	7.3	7.8	10.0	11.5	11.8	10.9	(37)
(38)		5% WB	MCD BR	12.5	14.0	18.0	19.4	17.6	16.3	16.8	17.4	20.1	19.3	17.6	13.5	(38)
(39)			MCWBR	10.3	10.8	11.1	10.5	8.8	8.0	7.2	7.1	9.2	10.9	11.5	10.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.263	0.278	0.321	0.342	0.355	0.363	0.372	0.325	0.290	0.262	0.245	(40)	
(41)		taud	2.486	2.511	2.547	2.425	2.416	2.428	2.420	2.365	2.467	2.560	2.542	2.518	(41)	
(42)		Ebn at Noon	851	921	956	932	919	907	893	868	884	874	834	822	(42)	
(43)		Edh at Noon	60	74	84	106	111	111	110	112	91	70	57	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.48	2.49	3.73	4.94	5.88	6.48	6.95	5.69	4.17	2.65	1.60	1.13	(44)	
(45)		RadStd	0.12	0.18	0.25	0.25	0.48	0.39	0.30	0.38	0.36	0.25	0.18	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page